

AWS Observability

1. Introduction

Observability is the ability to understand the internal state and behavior of a system by analyzing the data it produces. In AWS environments, observability is built around three primary sources of information: **metrics, logs, and traces**.

Modern applications often use distributed architectures involving microservices, containers, serverless functions, databases, APIs, and multiple AWS services. When an application fails or becomes slow, simply knowing that something is wrong is not enough. Engineers need to determine **what happened, where it happened, and why it happened**.

AWS provides a comprehensive set of observability services that help teams:

- Monitor infrastructure and application performance
- Collect and analyze logs
- Track requests across distributed applications
- Detect failures and performance issues
- Create dashboards for real-time visibility
- Configure alerts and automated responses
- Investigate security and operational events
- Identify AWS service disruptions
- Improve reliability and optimize cloud resources

Together, these services provide visibility across the entire AWS environment and support effective **monitoring, troubleshooting, incident response, and operational management**.

2. Major AWS Observability Services

2.1 Amazon CloudWatch

Amazon CloudWatch is the central monitoring and observability service in AWS. It collects and monitors metrics, logs, events, and other operational data from AWS resources and applications.

CloudWatch can monitor services such as:

- Amazon EC2
- Amazon RDS
- AWS Lambda
- Amazon ECS
- Amazon EKS
- Application Load Balancers
- Amazon S3
- Custom applications and workloads

CloudWatch provides capabilities for:

- **Metrics:** Monitor numerical measurements such as CPU utilization, memory usage, request count, latency, and error rates.
- **Logs:** Collect and analyze application and infrastructure logs.
- **Alarms:** Trigger notifications or automated actions when defined conditions are met.
- **Dashboards:** Visualize system health and performance in a centralized view.
- **Logs Insights:** Query and analyze large volumes of log data using a purpose-built query language.
- **Events and automation:** Respond to changes in AWS resources and application activity.

CloudWatch acts as a foundational observability service for many AWS workloads.

2.2 AWS CloudTrail

AWS CloudTrail provides auditing and governance visibility by recording API activity within an AWS account.

It helps answer important security and operational questions such as:

- Who performed an action?
- What action was performed?
- When did it happen?

- Which AWS resource was affected?
- Where did the request originate?

For example, if an important AWS resource is unexpectedly deleted or modified, CloudTrail can help identify the API activity responsible for the change.

CloudTrail is particularly important for:

- Security investigations
- Compliance and auditing
- Governance
- Incident investigation
- Tracking configuration changes

CloudTrail events can be stored in **Amazon S3** for long-term retention and can also be integrated with **CloudWatch Logs** for monitoring and alerting.

Key distinction: CloudTrail focuses primarily on **API activity and account actions**, while CloudWatch focuses on **system and application monitoring**.

2.3 AWS X-Ray

AWS X-Ray provides distributed tracing for applications. It helps engineers understand how requests travel through different components of a distributed system.

For example, a single user request might travel through:

API Gateway → Lambda → Application Service → Database

If the request is slow, X-Ray can help identify which component is contributing most to the latency.

X-Ray provides information about:

- Request flow
- Service dependencies
- Latency
- Errors
- Faults
- Downstream service calls

It also provides a **service map**, which gives a visual representation of relationships between application components.

X-Ray is particularly useful for:

- Microservices architectures
 - Serverless applications
 - Distributed applications
 - Troubleshooting latency
 - Identifying bottlenecks
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2.4 Amazon CloudWatch Logs

CloudWatch Logs provides centralized log collection, storage, and analysis for AWS resources and applications.

Logs can be collected from sources such as:

- EC2 instances
- Lambda functions
- ECS workloads
- VPC Flow Logs
- Applications
- AWS services

CloudWatch Logs organizes log data using **log groups** and **log streams**.

Important capabilities include:

Log Groups and Streams

A **log group** is a collection of related log streams. A **log stream** represents a sequence of log events from a particular source.

Retention Policies

Organizations can configure how long logs should be retained, helping control storage costs and meet operational or compliance requirements.

Metric Filters

Metric filters can identify specific patterns in logs and convert them into CloudWatch metrics.

For example, an application log containing repeated ERROR messages can be used to generate a metric that triggers an alarm.

Subscription Filters

Subscription filters can forward log data to other services for processing or analysis, including Lambda and Amazon OpenSearch Service.

CloudWatch Logs therefore provides the foundation for **centralized logging in AWS environments**.

2.5 Amazon CloudWatch Alarms

CloudWatch Alarms monitor metrics or metric expressions and automatically respond when defined conditions are met.

For example, an alarm could monitor:

EC2 CPU Utilization > 80% for 5 minutes

When the condition is satisfied, the alarm can enter the **ALARM** state and trigger an action.

Possible actions include:

- Sending an Amazon SNS notification
- Triggering an automated scaling action
- Stopping or restarting an instance
- Initiating an automated response

CloudWatch Alarms support three primary states:

State	Meaning
OK	The monitored condition is within the configured threshold
ALARM	The configured threshold has been breached
INSUFFICIENT_DATA	There is not enough data to determine the current state

CloudWatch also supports **composite alarms**, allowing multiple alarm conditions to be combined into a single higher-level alert.

2.6 CloudWatch Dashboards

CloudWatch Dashboards provide a centralized visual representation of AWS resources and application performance.

Dashboards can contain widgets such as:

- Line graphs
- Number displays
- Bar charts
- Alarm status
- Log queries
- Metric visualizations

A DevOps or SRE team can create a dashboard containing important indicators such as:

- EC2 CPU utilization
- Application request count
- API latency
- Error rate
- Database connections
- Load balancer health
- Alarm status

This provides a **single operational view** of system health and reduces the need to inspect individual AWS services separately.

2.7 Amazon Managed Service for Prometheus

Amazon Managed Service for Prometheus (AMP) is a fully managed monitoring service compatible with the open-source **Prometheus** ecosystem.

It is particularly useful for **containerized and Kubernetes environments**, including Amazon EKS.

AMP supports **PromQL (Prometheus Query Language)** and allows organizations to collect, store, query, and monitor Prometheus-compatible metrics without managing the underlying Prometheus infrastructure themselves.

Key benefits include:

- Managed Prometheus infrastructure
- PromQL support
- Integration with Amazon EKS
- Scalable metrics collection
- Reduced operational overhead
- Integration with visualization tools such as Amazon Managed Grafana

AMP is especially suitable for teams already familiar with Prometheus-based monitoring.

2.8 Amazon Managed Grafana

Amazon Managed Grafana is a fully managed visualization and analytics service based on Grafana.

It can connect to multiple data sources, including:

- Amazon CloudWatch
- Amazon Managed Service for Prometheus
- Amazon OpenSearch Service
- Other supported data sources

Grafana allows teams to create customizable dashboards for infrastructure and application monitoring.

Common use cases include:

- Kubernetes monitoring
- Application performance monitoring
- Infrastructure dashboards
- Metrics visualization
- Alerting
- Multi-source observability

A common AWS observability architecture is:

Amazon Managed Prometheus → Amazon Managed Grafana

Prometheus collects and stores metrics, while Grafana provides dashboards and visualization.

2.9 AWS Health Dashboard

The **AWS Health Dashboard** provides information about the health of AWS services and events that may affect an AWS account or its resources.

It can provide information about:

- AWS service disruptions
- Scheduled maintenance
- Operational issues
- Account-specific events
- Resource-specific impact

This is useful when an application appears to be experiencing problems and engineers need to determine whether the issue originates from their infrastructure or from an underlying AWS service.

For example:

Application latency increases → Check application metrics → Check AWS Health Dashboard → Identify an AWS service event

This helps teams correlate application problems with AWS infrastructure events.

2.10 Amazon OpenSearch Service

Amazon OpenSearch Service provides search, analytics, and visualization capabilities for large volumes of operational and log data.

It is commonly integrated with CloudWatch Logs to provide advanced log analysis.

A typical flow is:

Application / AWS Service → CloudWatch Logs → OpenSearch → OpenSearch Dashboards

OpenSearch is useful when teams require capabilities such as:

- Full-text log search
- Advanced log analysis
- Large-scale log analytics
- Custom dashboards
- Visualization
- Correlation of events across multiple sources

Compared with basic log querying, OpenSearch can provide more advanced search and analytics capabilities for large-scale observability environments.

3. The Three Pillars of Observability

Observability is commonly organized around three primary pillars: **Metrics, Logs, and Traces**.

Pillar	AWS Services	Purpose
Metrics	CloudWatch Metrics, Amazon Managed Service for Prometheus	Numerical measurements collected over time, such as CPU utilization, latency, request count, and error rate
Logs	CloudWatch Logs, Amazon OpenSearch Service	Detailed records of events and application activity used for troubleshooting and analysis
Traces	AWS X-Ray	Tracks requests across distributed application components to identify latency, failures, and bottlenecks

These three pillars complement each other.

4. Supporting Observability Capabilities

In addition to the three primary pillars, AWS provides services that support auditing, visualization, alerting, and incident investigation.

Capability	AWS Service	Purpose
Monitoring	Amazon CloudWatch	Monitor infrastructure and applications
Logging	CloudWatch Logs	Centralize and analyze logs
Tracing	AWS X-Ray	Track distributed requests
Auditing	AWS CloudTrail	Record AWS API activity and account actions
Alerting	CloudWatch Alarms	Detect threshold violations and trigger responses
Visualization	CloudWatch Dashboards	Provide centralized monitoring views
Prometheus Monitoring	Amazon Managed Service for Prometheus	Managed Prometheus-compatible metrics monitoring
Visualization & Analytics	Amazon Managed Grafana	Build dashboards using multiple observability data sources
AWS Service Health	AWS Health Dashboard	Identify AWS service events affecting workloads
Advanced Log Analytics	Amazon OpenSearch Service	Search, analyze, and visualize large-scale log data

5. Why AWS Observability Matters

A strong observability strategy helps organizations move beyond simply detecting that a system is down. It provides the information required to understand **why** the problem occurred.

Key benefits include:

Faster Troubleshooting

Engineers can use metrics, logs, and traces together to identify the source of failures and performance problems.

Proactive Monitoring

Alarms and anomaly detection can identify unusual behavior before it becomes a major incident.

Improved Application Performance

Latency, error rates, resource utilization, and request flows can be analyzed to identify performance bottlenecks.

Better Incident Response

Centralized dashboards, logs, alerts, and traces give operations teams the information they need during incidents.

Security and Compliance

CloudTrail provides an audit trail of AWS API activity, supporting security investigations and compliance requirements.

Better Reliability

Continuous visibility into infrastructure and applications helps teams identify reliability issues and improve system availability.

Cost Awareness

Monitoring resource utilization can help identify over-provisioned or underutilized resources and support cloud cost optimization.
